The Hardness Of 1018 Steel, 2011-Aluminum, 360 Brass, and Ductile Cast Iron

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Performed 02/28/2024

Reported 03/14/2024

Date: 03/14/2024

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Introduction

The methods of hardness testing are explored in the Hardness of Metals Lab. Specifically, the Rockwell Hardness test and the Vickers Hardness test are used. The materials consist of 1018 Steel, 2011-T3 Aluminum, Cast Iron and 360 Brass. The property hardness is defined by resistance to localized plastic deformation. Often hardness is associated with strength of a material and is crucial when trying to cut another material. In short, a harder material can be cut, scratched, or indented by a softer material. The main operation of any hardness test is to impact the material being tested with some level of force. By analyzing the indent on the material, a hardness rating can be developed. Unfortunately, there is not a singular test that is best for every material which means that the test must be fitted to the material to get the best results. For most materials the Rockwell test is a reliable method if the appropriate scale is selected. According to the University of Maryland, for softer material it is best to use the "B" scale which uses lower loads and for harder material it is best to use the "C" scale.[1] In this experiment, the "C" and "G" scale is used for testing the 1018 Steel while the "B" scale is used for testing the aluminum and brass. The "A" scale, which is the hardest scale, is used for the Cast Iron. The Rockwell test scales from A to V with "A" being the hardest and "V" being the softest. The Rockwell hardness test works by applying an initial load to an indenter and penetrating the test material which is followed by an even greater load to make the indent on the test material.[2] The hardness is measured as a function of the change in height in the indent. The hardness number is then found by using equation 1 below.

$$HRA = E - \frac{(h_2 - h_1)}{.002} \qquad \text{Eqn.1 } [2]$$

The letter after "HR" indicates the test scale that was used, and "E" is a test constant that is applied to each test based on the indenter type. The two height variables are the initial load indent depth and the final load indent depth.[2]

Another test type is the Vickers Hardness test which uses a diamond indenter and is great for extremely hard materials and thin materials. The main difference between Rockwell and Vickers hardness test is the shape of the indenter. According to the University of Maryland, the Vickers hardness test uses a pyramid-shaped indenter rather than the hemispherical shape used in the Rockwell test. A great advantage of the Vickers test indenter is its ability to resist deformation over time due to being made from a diamond[1]. Consequently, this means that the Vickers test machine is much more expensive than the Rockwell test machine. The Vickers test is identified with "HV" and can use a variety of loads. For this experiment a load of 1kgf is used for 1018 Steel and Cast Iron while a load of 500gf is used for the 2011-T3 Aluminum and 360 Brass. The Vickers applies a load to the indenter and leaves an impression similar to that of the Rockwell hardiness test. The Hardiness number is found by measuring the two-dimensional lengths of the impression on the test material or the penetration width[2]. The average of these two lengths are taken and applies to equation 2 below.

$$HV = 1.854 \frac{P}{d^2} \qquad \text{Eq. } 2^2$$

The variable "P" is the load and "d" is the average penetration width that is found during the test.

The information gained from the hardness test has different scales and rating depending on the test that is used. To compare materials that undergo different hardiness tests the numbers must be converted to a number on the same scale. However, once the numbers are converted the information can be very useful in determining whether a material will cut or not. If an engineer needs to cut steel, then they need to know how hard that steel is so they can use a cutting tool that is harder than what they are trying to cut. Hardness can also give insight into the strengths of the material.

Conclusions

The main purpose of this experiment was to test the hardness of 1018 Steel, T6-2011 Aluminum, Cast Iron, and 360 Brass using the Rockwell and Vickers hardness test. It was concluded the 1018 Steel had the highest hardness withT6-2011 Aluminum being the softest material. Comparing values with the literature, the Rockwell hardness test proved to be more accurate with aluminum and brass while the Vickers hardness test was more accurate with steel and cast iron. This aligned with the theoretical advantage of the Vickers hardness test when dealing with harder materials. Across the board, all experimental values were higher or on the higher end of the theoretical hardness of each sample material. This could be because MatWeb tested their hardness using the Brinell Hardness test and this experiment used Rockwell Hardness test and then converted it to Brinell Hardness. When converting between unit systems mistakes were possible, especially when a chart was used. This experiment also only conducted 5 trials for each which is more than likely less than what MatWeb did. The standard deviation of the samples was very high, which can be seen as proof of error during testing.

To properly compare the results a common scale had to be chosen. The common scale that was used was the Brinell Hardness number. This is very important because depending on the scale chosen for the Rockwell Hardness test a lower number could signify a greater hardness than a higher number. For example, a hardness number of HRC20 is much greater than a hardness number of HRG70 despite it being a smaller number. When compared properly, the order from hardest to softest was 1018 Steel, Ductile Cast Iron, 360 Brass, and 2011-T3 Aluminum. (The aluminum was harder than the brass in the Vickers test but proved to be softer in the Rockwell test.) This order was identical to the yield stress order of the Tensile Test Experiment done on the same sample. Another interesting point was the extremely high hardness

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of the 1018 Steel. The 1018 Steel matched its extremely high yield strength with both properties being almost twice the expected value. The hardness number of a material was important because it dictated whether one material could be cut or penetrated by another material. The strength of a particular material was indicated by its hardness number. To avoid the possibility of damaging parts, engineers had to be careful when selecting materials for parts that interacted. A scratch from a hard material on a soft material could lead to fracture failure when a crack was propagated over time. Hammers such as mallets were made with a rubber head to avoid damaging the surface when force was applied to it. For example, oil was used by combustion engines to avoid having two parts damaged, when they scratched and rubbed against each other. In Subtractive Manufacturing, the hardness of the tool was required to be greater than the workpiece precisely due to this concern. If the hardness of the tool, the workpiece, or the machine were not known, any of these pieces could break and could even leave the user with an injury.

References

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